

HW# 15

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Problem 1:

matrix of rotation is:

$$R_{xy} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$v = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$$

$$u = R_{xy} v \\ = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$$

$$u = \begin{bmatrix} (-3+\sqrt{3})/2 \\ (3\sqrt{3}+1)/2 \\ 2 \end{bmatrix} \Rightarrow u \text{ a rotated matrix.}$$

Problem 2:

reflection:- $y = \frac{4}{3}x \Rightarrow \begin{bmatrix} 1 \\ 4/3 \end{bmatrix} x$

$$y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$u = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow u = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$\begin{aligned}
 & 2UU^T - I \\
 &= 2 \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 18/25 & 24/25 \\ 24/25 & 32/25 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 R &= \begin{bmatrix} -7/25 & 24/25 \\ 24/25 & 7/25 \end{bmatrix}
 \end{aligned}$$

Stretching :-

$$\begin{aligned}
 x &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} & y &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\
 S &= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}
 \end{aligned}$$

Rotation :-

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \Rightarrow R = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Composition :-

$$RS = \begin{bmatrix} -7/25 & 24/25 \\ 24/25 & 7/25 \end{bmatrix}$$

$$\begin{aligned}
 \gamma SR &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -7/25 & 24/25 \\ 24/25 & 7/25 \end{bmatrix} \\
 &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -14/25 & 48/25 \\ 72/25 & 21/25 \end{bmatrix} \\
 &= \begin{bmatrix} 14/25 & -48/25 \\ -72/25 & -21/25 \end{bmatrix}
 \end{aligned}$$

Problem 3:

$$a) R_v^0 = \begin{bmatrix} \cos\theta & 0 & 0 \\ 0 & \cos\theta & 0 \\ 0 & 0 & \cos\theta \end{bmatrix} + \begin{bmatrix} (1-\cos\theta) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sin\theta \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & 0 & 0 \\ 0 & \cos\theta & 0 \\ 0 & 0 & \cos\theta \end{bmatrix} + \begin{bmatrix} 1-\cos\theta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\sin\theta \\ 0 & \sin\theta & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & 0 \\ 0 & 0 & \cos\theta \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\sin\theta \\ 0 & \sin\theta & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

It is a rotation around x axis from +ive y to z axis at the angle θ .

$$R_u^{7/4} = \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix} + \frac{2-\sqrt{2}}{2} \begin{bmatrix} 1/9 & 2/9 & 2/9 \\ 2/9 & 4/9 & 4/9 \\ 2/9 & 4/9 & 4/9 \end{bmatrix}$$

$$+ \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -2/3 & 2/3 \\ 2/3 & 0 & -1/3 \\ -2/3 & 1/3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix} + \frac{1}{18} \begin{bmatrix} 2-\sqrt{2} & 4-2\sqrt{2} & 4-2\sqrt{2} \\ 4-2\sqrt{2} & 8-4\sqrt{2} & 8-4\sqrt{2} \\ 4-2\sqrt{2} & 8-4\sqrt{2} & 8-4\sqrt{2} \end{bmatrix}$$

$$+ \begin{bmatrix} 0 & -\sqrt{2}/3 & \sqrt{2}/3 \\ \sqrt{2}/3 & 0 & -1/\sqrt{2}(3) \\ -\sqrt{2}/3 & 1/3\sqrt{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix} + \begin{bmatrix} \frac{2-\sqrt{2}}{18} & \frac{4-2\sqrt{2}}{18} & \frac{4-2\sqrt{2}}{18} \\ \frac{4-2\sqrt{2}}{18} & \frac{8-4\sqrt{2}}{18} & \frac{8-4\sqrt{2}}{18} \\ \frac{4-2\sqrt{2}}{18} & \frac{8-4\sqrt{2}}{18} & \frac{8-4\sqrt{2}}{18} \end{bmatrix}$$

$$+ \begin{bmatrix} 0 & -\sqrt{2}/3 & \sqrt{2}/3 \\ \sqrt{2}/3 & 0 & -1/3\sqrt{2} \\ -\sqrt{2}/3 & 1/3\sqrt{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix} + \begin{bmatrix} \frac{2-\sqrt{2}}{18} & \frac{2-4\sqrt{2}}{9} & \frac{2+2\sqrt{2}}{9} \\ \frac{2+2\sqrt{2}}{9} & \frac{8-4\sqrt{2}}{18} & \frac{8-7\sqrt{2}}{18} \\ \frac{2-4\sqrt{2}}{9} & \frac{8-\sqrt{2}}{18} & \frac{8-4\sqrt{2}}{18} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1+4\sqrt{2}}{9} & \frac{2-4\sqrt{2}}{9} & \frac{2+2\sqrt{2}}{9} \\ \frac{2\sqrt{2}}{9} & \frac{8+5\sqrt{2}}{18} & \frac{8-7\sqrt{2}}{18} \\ \frac{2-4\sqrt{2}}{9} & \frac{8-\sqrt{2}}{18} & \frac{8+5\sqrt{2}}{18} \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$R_u^{45}(v) = \begin{bmatrix} \frac{3+2\sqrt{2}}{3} \\ \frac{12+5\sqrt{2}}{6} \\ \frac{12-7\sqrt{2}}{6} \end{bmatrix}$$

$$c) R_{y2}^{\pi/3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{bmatrix}$$

$$R_{xy}^{\pi/3} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{y2}^{\pi/3} R_{xy}^{\pi/3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/4 & 1/4 & -\sqrt{3}/2 \\ 3/4 & \sqrt{3}/4 & 1/2 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} & 0 \\ -\sqrt{3}/2 & 1/2 & -\sqrt{3} \\ 3/2 & \sqrt{3}/2 & 1 \end{bmatrix}$$